

The Process-Safety Teardown Checklist

Twenty-one checks drawn from seven investigated disasters.
Run them against your own site this week.

01

Givaudan Sense Colour

LOUISVILLE, KY · 12 NOV 2024 · 2 FATALITIES · CSB 2024-06-I-KY

The harmless-looking material nobody thought to reactivity-test.

- ☐ Reactivity-test your low-hazard materials at real operating and upset conditions.
- ☐ For anything that could run away, what scenario was the relief actually sized for?
- ☐ Does your PSM effort chase the scary-sounding chemicals, or the truly reactive ones?

02

Formosa Plastics

ILLIOPOLIS, IL · 23 APR 2004 · 5 FATALITIES · CSB 2004-10-I-IL

One bypassable procedure standing in for a real barrier.

- ☐ Count the independent protection layers on your top three scenarios. One operator is not one.
- ☐ Every bypassable interlock: is the bypass managed under MOC, or waved through by phone?
- ☐ A near-miss at a sister site is your finding too, not “not applicable here.”

03

Bayer CropScience

INSTITUTE, WV · 28 AUG 2008 · 2 FATALITIES · CSB 2008-08-I-WV

A startup pushed before the unit was ready to run.

- ☐ Your last post-turnaround pre-startup review: performed and closed, or just signed?
- ☐ New control interface: were operators proven competent before go-live, not during it?
- ☐ Is the hardest, least-familiar work running on 60-hour weeks and 18-hour shifts?

04

BP Texas City Refinery

TEXAS CITY, TX · 23 MAR 2005 · 15 FATALITIES · CSB 2005-04-I-TX

A low injury rate mistaken for a healthy process.

- ☐ Split your safety numbers. Can you name one process-safety metric you watch weekly?
- ☐ Your critical gauges: what is each one calibrated to see, and what is it blind to?
- ☐ Any relief or PHA study years overdue? That is a decision, not a backlog item.

05

Esso Longford Gas Plant

LONGFORD, VICTORIA · 25 SEP 1998 · 2 FATALITIES · ROYAL COMMISSION

The people who understood the hazard had been moved off-site.

- ☐ Can your shift read the early sign of your worst failure mode, or was it never taught?
- ☐ Did your last engineering reorg get an MOC review, or pass as an org-chart change?
- ☐ Near-miss log and HAZOP schedule: is your worst unit's study done, or just planned?

06

DuPont La Porte

LA PORTE, TX · 15 NOV 2014 · 4 FATALITIES · CSB 2015-01-I-TX

Every barrier already gone before the last valve was opened.

- ☐ Piping moved to clear a plug or restore flow: did that change ever get an MOC?
- ☐ Is your toxic-gas detection sized for toxicity, or borrowed from a flammability limit far above it?
- ☐ Count last month's alarms your crews treated as normal. That count is normalized deviance.

07

Aghorn Operating

ODESSA, TX · 26 OCT 2019 · 2 FATALITIES · CSB 2020-01-I-TX

A safety system that lived in one man's memory.

- ☐ Which alarms send a person into the hazard just to learn what the alarm means?
- ☐ Pull the calibration records for your toxic-gas detection. No records, no detection.
- ☐ Any energy-isolation step that is not written down is a program running on memory.

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